Threat Detection & Attack Mapping using MITRE ATT&CK and Cyber Kill Chain

1. Objective

The objective of this project is to detect and analyze suspicious authentication activity using SIEM logs and map the identified attack to industry-standard frameworks such as MITRE ATT&CK and the Cyber Kill Chain.

2. Data Source

The analysis was performed using Windows Security Logs ingested into Splunk SIEM.

- Event ID Used: **4625 (Failed Login Attempts)**
- Log Source: Windows Event Logs
- Environment: Simulated attack using brute-force attempts over RDP

3. Detection

Multiple failed login attempts were identified from a single source IP within a short time frame. This behavior indicates a potential brute-force attack targeting system authentication.

4. Investigation

Key Findings:

- **Attacker IP:** 192.168.110.129
- **Targeted Accounts:** administrator, patil
- **Attack Pattern:** High-frequency automated login attempts
- **Timeline Insight:** Sudden spike in login attempts indicating automated attack

Analysis:

- A single IP generated the majority of failed login attempts
- Privileged accounts (administrator) were primarily targeted
- Activity pattern suggests automated password guessing attack

5. MITRE ATT&CK Mapping

- **T1110 – Brute Force**
  Multiple failed login attempts indicate password guessing attack
- **T1021 – Remote Services (RDP)**
  Attack was performed via Remote Desktop Protocol (RDP)

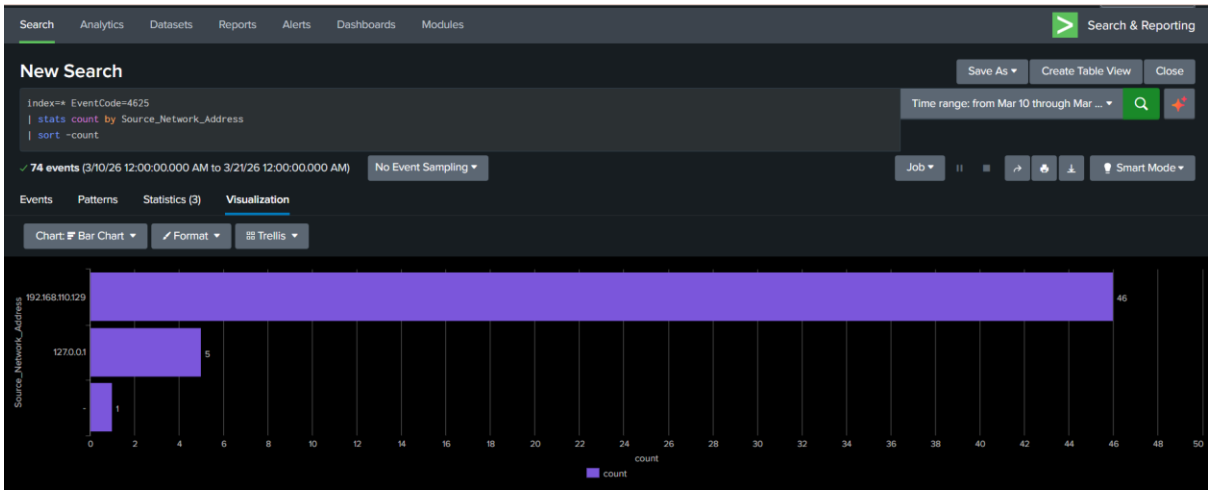
6. Cyber Kill Chain Mapping

The attack was analyzed using the Cyber Kill Chain model:

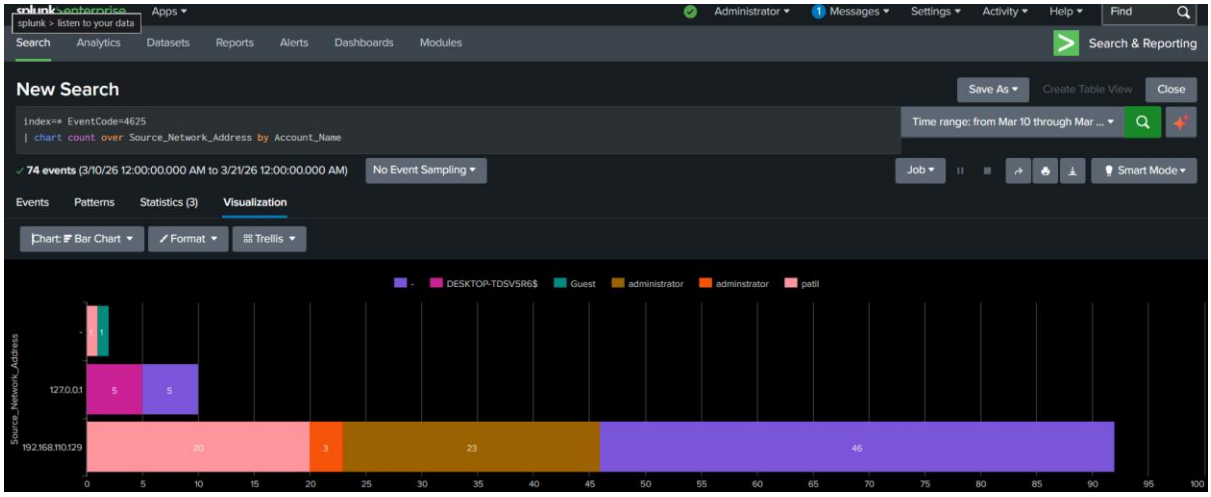
- **Reconnaissance:** Attacker identifies open RDP service
- **Weaponization:** Use of brute-force tool to automate login attempts
- **Delivery:** Login attempts sent to target system
- **Exploitation:** Password guessing attempts
- **Installation:** Not observed (no confirmed successful login)
- **Command & Control:** Not observed
- **Actions on Objectives:** Attempted unauthorized access

7. Visualization Insights

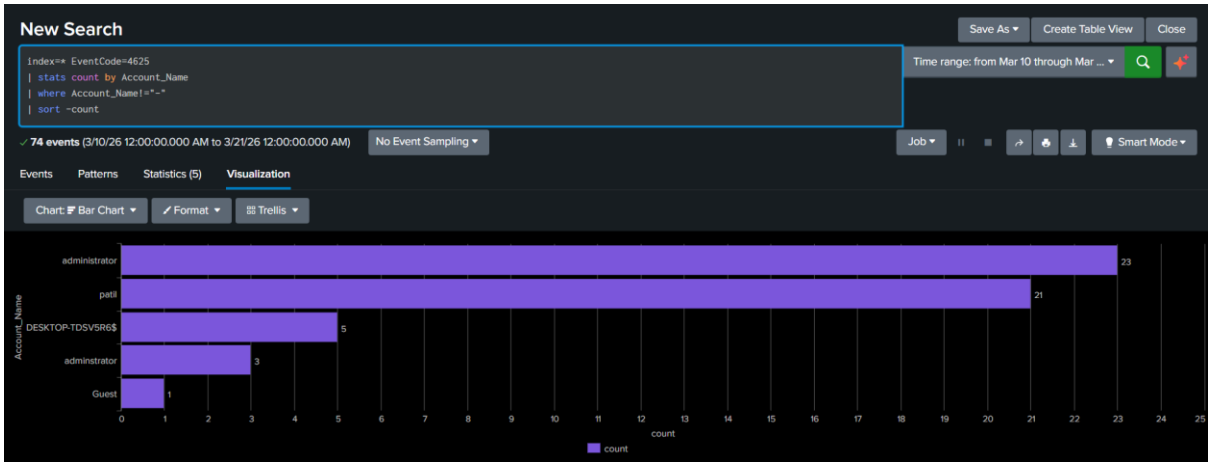
Shows spike in failed login attempts indicating automated brute-force activity.



Identifies the source IP responsible for the majority of failed login attempts.



Highlights user accounts targeted during the attack, mainly privileged accounts.



Shows relationship between attacker IP and targeted accounts to understand attack pattern.

This analysis demonstrates how SIEM data can be leveraged to detect, investigate, and map real-world attack techniques using industry frameworks.